

file I/O

2024年9月24日

10:53

stream: sequence of characters

stream I/O operations >>/<< : convert input/output data of the required type from/to a sequence of characters.

The input operator >> always skip whitespaces – spaces, tabs, newlines, formfeeds, carriage returns – before reading the next datum.

The **stream objects** of various **stream classes** also support the following **class member functions**:

- **open(const char* filename)**: create a stream and associate it with a **file** with the given **filename**.
- **close()**: close a stream created by **open()**.
- **eof()**: check if the **end of a file** is reached.
- **get(char& c)**: get the next character into the variable **c** from an **input stream**.
- **getline(char* s, int max-num-char, char terminator='\n')**: get a stream of characters and put it into the char array pointed by the variable **s**. **getline()** stops when either
 - (**max-num-char - 1**) characters are read; or,
 - the **stopping character** **terminator** ('\n' by default) is seen.

Notice that the stopping character is **not** read into the array, and the **null character** is automatically inserted at the end of **s**.

- **put(char c)**: put the character represented by the variable **c** onto an **output stream**.

iostream=istream+ostream=input stream+output stream

istream cin : **standard (or console) input**, by default, is the keyboard.

ostream cout : **standard (or console) output**, by default, is the screen.

ostream cerr : **standard (or console) error output**, by default, is the screen. From now on, you should send your **error messages** to **cerr** instead of **cout**.

fstream=ifstream+ofstream=input file stream+output file stream

(input file必须先于指令存在, output file存在则覆盖不存在则自动生成)